Loss Obeys a Log-Share Law, Downstream Accuracy Does Not: A Dose–Response Study of Pre-Training Data Mixtures

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Abstract

The composition of a pre-training corpus is chosen mostly by heuristics. Prior data mixing laws predict validation loss from domain proportions, but they do not tell us whether downstream accuracy follows the same shape. We test a concrete hypothesis: at a fixed model size and token budget, performance is log-linear in each domain's share and saturates near 20%, code is the strongest driver of reasoning scores ($r > 0.7$), and more than 50% web text hurts knowledge tasks. We train 81 small GPT models (17M parameters, 60M tokens each) from scratch on 53 controlled mixtures of web text, Wikipedia, code, and scientific papers, with seed replicates at 14 mixtures. The hypothesis holds for language-modeling loss and essentially nowhere else. Held-out loss on a domain is log-linear in that domain's share (R^2 of 0.98–0.9996, against 0.67–0.87 for a linear fit), but it does not saturate at 20%: loss keeps falling by 0.36–0.55 nats between 20% and 100%. Downstream accuracy is nearly flat across mixtures, with most differences inside seed noise. Code share does not correlate with our reasoning composite ($r = -0.10$ on 16 random mixtures), and 75% web text is indistinguishable from 25–50% on knowledge probes. The one robust downstream effect is that a pure-code corpus costs 7.4 points of fine-tuned question-answering accuracy. At this scale, a log-share law fitted on about thirty runs predicts loss on unseen mixtures, but these loss differences do not turn into measurable downstream differences.

1 Introduction

How much web text, code, encyclopedic text, and scientific writing should a pre-training corpus contain? This choice is one of the few levers that changes what a language model can do at a fixed compute budget, yet practitioners mostly set it by heuristics. A simple, predictable law linking domain proportions to performance would let us choose a mixture from a handful of small runs instead of trial and error at scale.

Recent work shows that such laws exist for validation loss. Ye et al. [2025] fit an exponential function of the mixture proportions to validation loss, and Shukor et al. [2025] extend scaling laws to include domain weights. RegMix [Liu et al., 2025] regresses a loss target on random mixtures and reports that web text correlates best with downstream performance, while Aryabumi et al. [2025] show that adding code to pre-training helps reasoning tasks.

However, these studies leave the *shape* of the relationship between a domain's share and downstream performance open. Loss-level laws do not tell us whether task accuracy follows the same curve, and the code result rests on a handful of mixtures, not on a dose–response curve. Folk beliefs about mixtures are also more specific than what has been tested: that returns from a domain diminish quickly, that code drives reasoning, and that too much web text dilutes knowledge.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Our main question is whether these beliefs survive a controlled test. We turn them into four claims. (H1) Performance is log-linear in a domain's share. (H2) Returns saturate at about 20% for any single domain. (H3) Code share is the strongest positive correlate of reasoning scores, with $r > 0.7$. (H4) Web text above 50% degrades knowledge-intensive tasks.

We test these claims with a dose–response study. We train 81 GPT models with 17M parameters from scratch, each on 60M tokens drawn from a controlled mixture of four domains: web text, Wikipedia, Python code, and arXiv papers. The design has three parts (section 3): single-domain sweeps that vary one share from 0 to 1, 16 random mixtures that give an unbiased sample for correlations and a held-out test set for mixture laws, and three seeds at 14 mixtures that give a noise floor. We evaluate every model on held-out loss per domain, on zero-shot and fine-tuned knowledge question answering, and on fine-tuned reasoning probes.

The hypothesis holds for loss and essentially nowhere else. In-domain loss is log-linear in the domain's share: the log law has 2.3 to 22 times lower leave-one-out error than a linear law, with R^2 between 0.98 and 0.9996. But there is no saturation at 20%; loss falls a further 0.36–0.55 nats from 20% to 100%, which is 59–98 seed standard deviations. Downstream accuracy is almost flat. Code share correlates with the reasoning composite at $r = -0.10$ on the random mixtures (95% CI $[-0.55, 0.57]$), and 75% web text is indistinguishable from 25% or 50% on both knowledge measures. The largest downstream effect is in the opposite direction from H3: a pure-code corpus loses 7.4 points of fine-tuned SciQ accuracy.

These results come with a clear boundary. Our models are four to five orders of magnitude smaller than production language models, and their zero-shot accuracy is 2–3 points above chance. The downstream nulls therefore show that the effects are small relative to our noise at this scale, not that they are absent at large scale.

In summary, our contributions are:

- We design a dose–response study of pre-training mixtures with 81 models on 53 mixtures, combining single-domain sweeps, random mixtures, and seed replicates so that each claim has a test and a noise estimate.
- We show that in-domain loss follows a log-share law with no saturation at 20%, and that the first 2% of a domain delivers 43–67% of its total loss benefit.
- We show that a log-share law fitted on 29 mixtures predicts loss on 16 unseen mixtures with correlation 0.94–0.9996, and we compare it with the exponential law of Ye et al. [2025].
- We find that downstream accuracy does not follow the hypothesized laws at this scale: it is flat within seed noise across interior mixtures, and neither the code–reasoning claim nor the web–knowledge claim is supported.

We review related work in section 2, describe the setup in section 3, report results in section 4, and discuss interpretation and limitations in section 5.

2 Related Work

Mixing laws for loss. Ye et al. [2025] propose data mixing laws: validation loss is a predictable function of the mixture proportions, exponential in the proportions, which can be fitted on a few sampled mixtures and nested with step and model-size scaling laws to choose a mixture for a larger model. Shukor et al. [2025] extend scaling laws to predict loss as a joint function of model size, token count, and domain weights. Both target validation loss. We use the same kind of small-run fitting, but we ask whether downstream accuracy follows similar shapes, and we compare the exponential form against a log-share form on held-out mixtures.

Choosing mixtures with proxy models. RegMix [Liu et al., 2025] trains many tiny models on random mixtures, fits a regression from mixture to a target loss, and transfers the best mixture to larger models. It reports that web corpora, and not sources usually seen as higher quality such as Wikipedia, correlate most positively with downstream performance. DoReMi [Xie et al., 2023] and DoGE [Fan et al., 2024] learn domain weights with a proxy model. These methods search for a good mixture. Unlike them, we do not optimize a mixture; we characterize the shape of the dose–response curve along each domain and test a specific threshold for web text. RegMix's finding on web text runs against the claim that heavy web text is harmful, which makes H4 a real test.

Code in pre-training. Aryabumi et al. [2025] run ablations on models with 470M to 2.8B parameters and report relative gains of up to 8.2% on natural-language reasoning and 4.2% on world knowledge when code is added to text-only pre-training. This result motivates H3. It uses a few mixtures, so it does not show how the benefit changes with the amount of code. We sweep code share over nine levels and measure its correlation with reasoning probes on random mixtures.

Scaling laws and small-model evaluation. Scaling laws describe loss as a power law in data and parameters [Kaplan et al., 2020, Hoffmann et al., 2022]. At a fixed budget a domain's share is its data quantity, so a logarithmic dependence of in-domain loss on share is the natural prior for H1. Work on small language models, such as the BabyLM challenge [Warstadt et al., 2023], evaluates mostly with likelihood-based and fine-tuning protocols because zero-shot benchmark accuracy is near chance at that scale. We follow this practice and measure reasoning by fine-tuning transfer.

3 Methodology

3.1 Problem formulation

Let $\boldsymbol{p} = (p_{\text{web}}, p_{\text{wiki}}, p_{\text{code}}, p_{\text{arxiv}})$ be a mixture on the simplex, where p_d is the fraction of training tokens drawn from domain d. We fix the architecture, tokenizer, token budget, optimizer, and schedule, and train one model per mixture. We then ask how a score $y(\boldsymbol{p})$, either held-out loss on a domain or accuracy on a downstream task, depends on $\boldsymbol{p}$.

Curve shapes along a sweep. For a single-domain sweep we compare four forms for y as a function of the swept share p:

$$\text{linear:} \quad y = a + b\,p, \tag{1}$$

$$\text{log:} \quad y = a + b \log p, \tag{2}$$

$$\text{exponential:} \quad y = c + k\,e^{tp}, \tag{3}$$

$$\text{log with plateau:} \quad y = a + b \log \min(p, p^\star). \tag{4}$$

The exponential form is the one-dimensional version of the law of Ye et al. [2025], and $p^\star$ is a fitted saturation point. H1 says that the log form beats the linear form. H2 says that $p^\star \approx 0.2$.

Laws over the whole simplex. To predict unseen mixtures we compare a linear (Scheffé) mixture model $y = \sum_j \beta_j p_j$, the log law in the domain's own share (equation 2), the multi-domain exponential law $y = c + k \exp(\sum_j t_j p_j)$ of Ye et al. [2025], and an additive log-share law

$$L_d(\boldsymbol{p}) = a_d + \sum_j b_{dj} \log p_j, \tag{5}$$

where L_d is held-out loss on domain d. The coefficients b_{dj} form a transfer matrix: b_{dj} is the change in loss on domain d per e-fold increase in the share of domain j.

3.2 Pre-training

Domains. We use four domains: WEB is English C4 [Raffel et al., 2020]; WIKI is WikiText-103 [Merity et al., 2017]; CODE is the Python corpus `codeparrot-clean`; ARXIV is the article bodies of the arXiv summarization dataset [Cohan et al., 2018]. We equalize the pools at 128.9M training tokens per domain and hold out separate documents for evaluation. One byte-level BPE tokenizer with 16,384 tokens, trained on a balanced sample, is shared by all runs.

Model and budget. Every model is a decoder-only transformer with 6 layers, width 384, 6 heads, context length 256, and tied embeddings, for 17.0M parameters. Every run sees exactly 60M tokens (3,662 steps of 64 sequences of 256 tokens). We train with AdamW [Loshchilov and Hutter, 2019] at learning rate 10^{-3}, 3% warm-up, cosine decay to 10% of the peak, weight decay 0.1, $\beta = (0.9, 0.95)$, gradient clipping at 1.0, and bfloat16 autocast. Each training sequence is drawn from domain d with probability p_d. We chose this size because the GPUs were shared with other jobs; section 5.1 and Appendix A describe this and other deviations from the original plan.

3.3 Mixture design

We train 81 runs on 53 distinct mixtures (table 1).

Table 1: Mixture design. The uniform mixture lies on all four sweeps. Seed replicates use three seeds each and give 28 degrees of freedom for the noise estimate.

Component	Mixtures	Purpose
Single-domain sweeps	share $p \in \{0, 0.02, 0.05, 0.1, 0.2, 0.35, 0.5, 0.75, 1\}$ per domain; the other three split $1 - p$ equally	curve shape, saturation, web threshold (H1, H2, H4)
Random mixtures	16 draws from $\mathrm{Dirichlet}(1, 1, 1, 1)$	unbiased correlations (H3); held-out test set for laws
Seed replicates	uniform; every p=0 and p=1 model; WEB at 0.2, 0.5, 0.75; CODE at 0.2, 0.5 (14 mixtures)	noise floor

Why three components? The sweeps are the only way to see the shape of a curve and to locate a saturation point. But shares on a simplex are negatively correlated with one another, and a sweep confounds a domain's effect with the sweep structure, so we use Dirichlet draws for correlations. Without seed replicates, no claim about curve shape or degradation is interpretable.

3.4 Evaluation

Loss. We compute held-out loss in nats per token on each domain over non-overlapping windows (0.27–1.0M tokens per domain).

Knowledge, zero-shot (K_{zs}). We score multiple-choice questions by choosing the answer with the highest length-normalized log-likelihood after the prompt "`Question: ...\nAnswer:`". K_{zs} is the mean accuracy on SCIQ [Welbl et al., 2017] (13,679 items) and ARC-EASY [Clark et al., 2018] (5,197 items). We also report ARC-Challenge, OpenBookQA [Mihaylov et al., 2018], and CommonsenseQA [Talmor et al., 2019] but exclude them from the composite.

Knowledge, fine-tuned (K_{ft}). We fine-tune each checkpoint for 400 steps on SCIQ training answers without the supporting passage, then score multiple choice on the 2,000 validation and test items.

Reasoning (R). Models of this size cannot reason zero-shot, so we measure fine-tuning transfer. R is the mean exact-match accuracy on LISTOPS [Nangia and Bowman, 2018] (generated; 4,000 training and 2,000 test examples) and PROOFWRITER [Tafjord et al., 2021] (3,000 training and 1,500 test examples), each after 500 fine-tuning steps. We also report 3-digit arithmetic after 1,500 steps, outside the composite.

Fine-tuning uses learning rate 3×10^{-4}, loss on answer tokens only, and the same data order for every checkpoint. Re-running the evaluation of a checkpoint reproduces every number exactly.

How we chose the composites. We fixed task inclusion using only the uniform-mixture pilot and a randomly initialized control, before any mixture comparison. K_{zs} keeps the two zero-shot tasks that were above chance. K_{ft} is the fallback we planned in case zero-shot signal was weak. R keeps the two fine-tuned tasks on which pre-training helped. We exclude arithmetic because pre-training hurt it in the pilot.

3.5 Baselines and statistical analysis

Baselines. We compare against the uniform mixture, pure-domain models ($p = 1$), leave-one-domain-out models ($p = 0$), a randomly initialized model for all fine-tuned probes, and chance for multiple choice. For curve shape the linear fit is the null.

Tests. For curve shape we use leave-one-out (LOO) RMSE as the primary criterion and AICc as a check, fitting on the nine shares above zero. For correlations we report Pearson r with bootstrap 95% confidence intervals and a Holm correction across the four domains. For degradation we use Welch t-tests across seeds. The noise floor is the pooled standard deviation across seeds at the 14 replicated mixtures. We fit simplex laws on the 29 structured mixtures in which every share is positive and test on the 16 random mixtures.

Table 2: Signal against noise. Loss varies across mixtures over 100 times more than across seeds. Downstream accuracy varies across mixtures about as much as across seeds. In-domain loss covers runs that contain the domain.

Metric	Range across mixtures	SD across mixtures	SD across seeds	Ratio
In-domain loss (nats)	0.7–1.5	–	0.004–0.008	> 100
K_{zs} (%)	26.0–28.7	0.39	0.26	1.5
K_{ft} (%)	39.0–49.1	1.35	0.55	2.4
LISTOPS (%)	35.5–42.2	1.03	0.76	1.3
PROOFWRITER (%)	56.0–70.6	2.70	2.58	1.0
R (%)	47.5–54.8	1.41	1.29	1.1
Arithmetic (%)	7.7–98.5	25.6	23.4	1.1

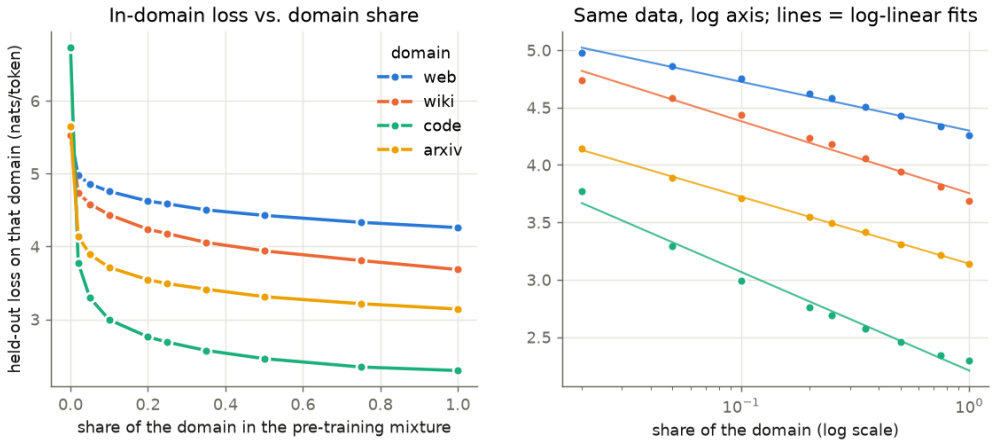


Figure 1: Held-out loss on each domain as its share of the mixture varies. *(Left)*Linear share axis. *(Right)*The same points on a log axis with log-linear fits. Loss is close to a straight line in log share over the whole range from 2% to 100%, with no bend at 20%.

4 Results

4.1 How much signal is there?

Table 2 sets the scale for everything that follows. We measure loss two orders of magnitude more precisely than it varies. Downstream accuracy is different: only the knowledge measures rise clearly above seed noise, and a few extreme mixtures drive that. Pre-training does help the probes. The randomly initialized control scores 37.6% on K_{ft}, 30.1% on LISTOPS, and 55.9% on PROOFWRITER, and the uniform mixture adds 9.3, 9.0, and 3.7 points.

4.2 H1: in-domain loss is log-linear in the domain's share

Figure 1 and table 3 show that H1 holds for loss. The two-parameter log law beats the linear law in every domain, with 2.3 to 22 times lower LOO error, and it is the best or near-best model everywhere. AICc agrees: the log law is 17–57 units below the linear law.

The law is a good approximation, not an exact description. Residuals from the log law are 0.01–0.10 nats, which is still larger than seed noise (0.004–0.008) for WEB, WIKI, and CODE. For ARXIV the fit is essentially exact.

Share 0 is a cliff, not a point on the curve. The first 2% of a domain delivers 43–67% of the total improvement from 0% to 100%. No smooth law fitted to $p > 0$ extrapolates to $p = 0$.

Table 3: Curve-shape comparison for in-domain loss, fitted on the nine shares above zero. LOO is leave-one-out RMSE in nats per token; lower is better and the best is in **bold**. "Exp" is the one-dimensional form of Ye et al. [2025].

	Loss at share				R^2		LOO error			
Domain	0	0.02	0.2	1.0	Linear	Log	Linear	Log	Exp	Log+plateau
WEB	5.53	4.97	4.62	4.26	0.863	0.984	0.120	0.044	**0.041**	0.051
WIKI	5.52	4.74	4.24	3.68	0.872	0.980	0.171	0.074	**0.056**	0.084
CODE	6.73	3.77	2.76	2.30	0.669	0.984	0.369	0.095	0.163	**0.074**
ARXIV	5.65	4.14	3.55	3.14	0.769	0.9996	0.208	**0.009**	0.092	0.025

Table 4: Saturation analysis for in-domain loss. A plateau starting at 1.00 means the fit found no saturation inside the sweep.

	% of 0→100% gain reached at			Further gain 20%→100%		
Domain	5%	20%	50%	nats	seed SDs	Fitted plateau start
WEB	53	71	87	0.36	74	1.00 (none)
WIKI	51	70	86	0.55	79	1.00 (none)
CODE	78	90	96	0.46	59	0.65
ARXIV	70	84	93	0.40	98	1.00 (none)

4.3 H2: no saturation at 20%

Returns diminish in the sense that any logarithm does. Each doubling of a domain's share buys a roughly constant 0.15 (WEB), 0.20 (WIKI), 0.32 (CODE), or 0.18 (ARXIV) nats. But nothing special happens at 20% (table 4). Loss falls a further 0.36–0.55 nats between 20% and 100%, which is 59–98 seed standard deviations. A log law with a plateau fits worse out of sample than a plain log law in three of four domains. Where the data prefer a plateau, for CODE, it starts at about 65%.

Downstream tasks have no rising segment to saturate. Between shares of 2% and 50%, K_{zs}, K_{ft}, and R stay inside the ± 2 SD seed band for every domain (figure 2). The log law is also not the preferred shape for downstream scores. A linear fit has the lowest LOO error in 8 of 12 downstream domain–metric pairs, because the only real effects sit at the high-share end: more WIKI raises knowledge scores and more CODE lowers them. This is the opposite of diminishing returns. In most of these fits R^2 is below 0.25, so there is little to explain.

4.4 H3: code is not a strong driver of reasoning scores

The hypothesized $r > 0.7$ is far from what we observe (figure 3, table 5). On the random mixtures, code share correlates with R at $r = -0.10$, and even the upper confidence limit of 0.57 is below 0.7. On all 53 mixtures $r = 0.01$. Removing code entirely changes nothing we can measure: R is 49.4% without code and 49.4% at uniform.

Does code help any reasoning task? Code has the largest positive association with one task, LISTOPS, which is the most code-like of our probes because its inputs are bracketed nested operators ($r = 0.39$, Holm-corrected $p = 0.014$). The effect is small. A mixture regression predicts 41.4% (SE 0.6) for a pure-code corpus against 38.6–39.3% for the other pure corpora, and the seed-replicated pure-code model is 1.6 points above uniform (Welch $p = 0.14$, 3 vs. 3 seeds).

4.5 H4: web text above 50% does not measurably hurt knowledge tasks

At 75% web, both knowledge measures are statistically indistinguishable from 50% or 25% (all $p > 0.27$; figure 4, table 6). The pure-web model is 1.2 points below uniform on K_{ft} (Welch $t = -3.7$, $p = 0.043$, uncorrected, 3 vs. 3 seeds) and unchanged on K_{zs}. We do not read much into this single result. We examined about ninety such contrasts, so it would not survive a multiplicity correction, and "all web" is also "no Wikipedia and no papers at all".

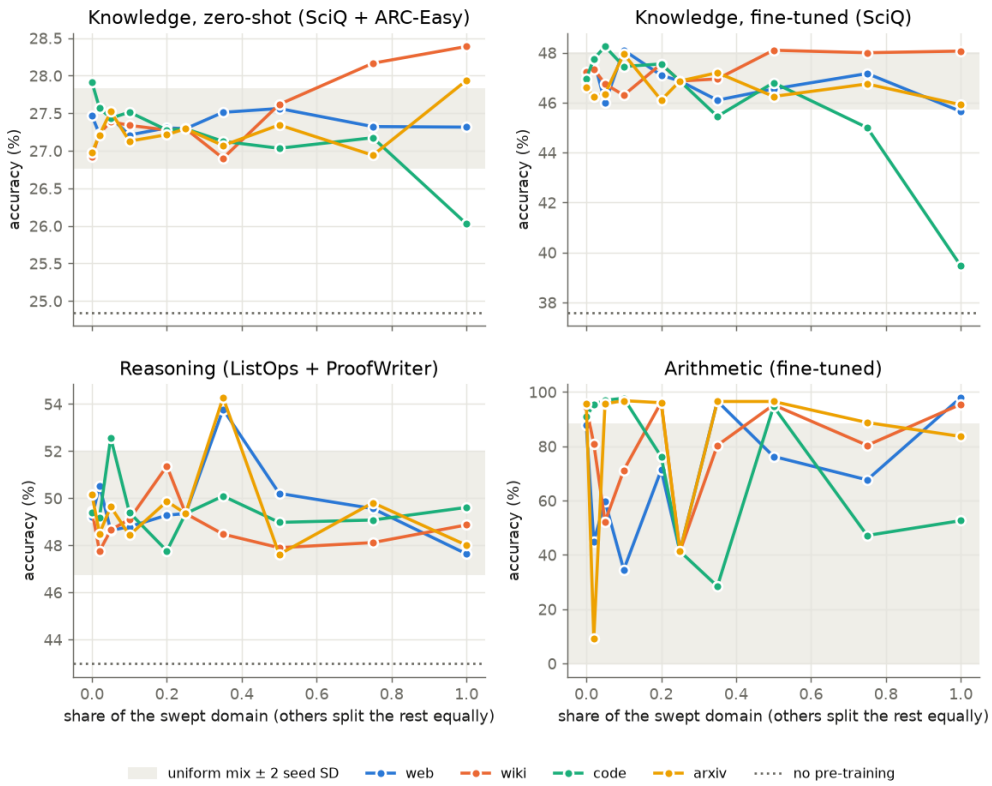


Figure 2: Downstream accuracy as each domain's share is swept. The grey band is the uniform-mixture mean ± 2 pooled seed SD. Points are single seeds except at replicated shares, where they are means of three. Between shares of 2% and 50% every score stays inside the band.

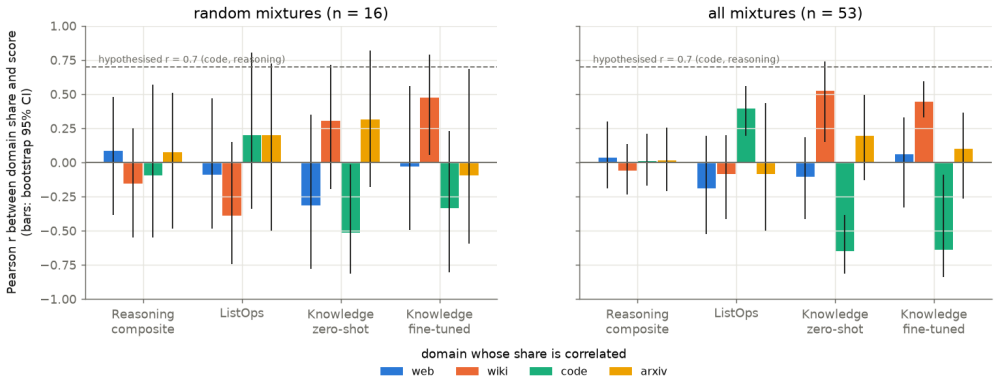


Figure 3: Pearson correlation between each domain's share and downstream scores, on the 16 random mixtures (the pre-specified test) and on all 53 mixtures. No domain correlates with the reasoning composite; code share is the strongest negative correlate of both knowledge measures.

Loss does respond to web share. Loss on WIKI rises from 4.20 to 4.28 and loss on ARXIV from 3.58 to 3.76 as web grows from 50% to 75%, and both collapse at 100%. The accuracy measures do not follow.

What does hurt knowledge scores? Code at very high shares. Pure code is 7.4 points below uniform on K_{ft} (39.5% vs. 46.9%, $t = -28$) and 1.3 points below on K_{zs}; 75% code is about 1.9 points below on K_{ft} (single seed). Pure Wikipedia is the best knowledge model, 1.2 points above uniform on K_{ft} and 1.1 on K_{zs} ($p = 0.05$ and 0.02, uncorrected).

Table 5: Correlation of code share with downstream scores, with bootstrap 95% confidence intervals where computed. **Bold** marks intervals that exclude zero. The last column names the domain with the strongest positive correlation on all mixtures.

Score	Random mixtures ($n = 16$)	All mixtures ($n = 53$)	Strongest domain
R	-0.10 $[-0.55, 0.57]$	0.01 $[-0.17, 0.21]$	none (all $\lvert r\rvert < 0.07$)
LISTOPS	0.20 $[-0.34, 0.81]$	$\mathbf{0.39}$ $[0.19, 0.56]$	CODE
PROOFWRITER	-0.20	-0.14	none significant
K_{zs}	-0.52	$\mathbf{-0.65}$ $[-0.82, -0.39]$	WIKI ($+0.52$)
K_{ft}	-0.34	$\mathbf{-0.64}$ $[-0.84, -0.09]$	WIKI ($+0.45$)

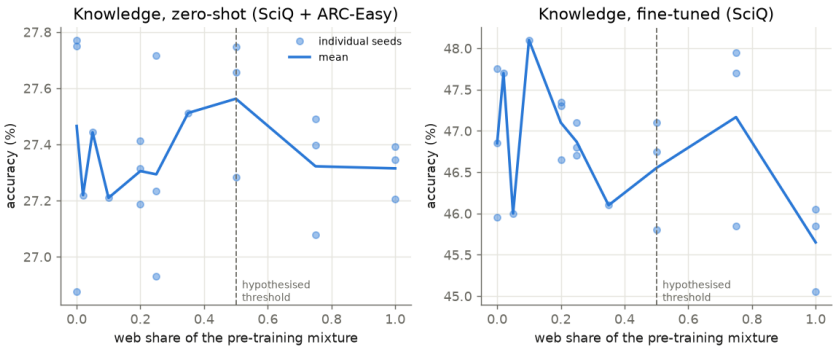


Figure 4: Knowledge accuracy along the web sweep. Dots are individual seeds. Neither measure drops between 50% and 75% web; only the pure-web model is lower, and only on K_{ft}.

4.6 Can a mixture law predict unseen mixtures?

Loss is predictable. A log-share law predicts loss on unseen mixtures with correlation 0.94–0.9996 (figure 5, table 7). No single form wins. The exponential law of Ye et al. [2025] is best where domains help each other (WEB and WIKI), and log-share laws extrapolate far better to tiny shares; the random set includes shares as small as 0.1%. The errors of all laws concentrate on the three random mixtures with a share below the fitted range of 2% (0.1% code, 0.6% and 1.2% arXiv).

Which domains help each other? The additive law gives an interpretable transfer matrix (table 8). Web and Wikipedia help each other at roughly a third of the own-domain rate. Loss on arXiv and on code depends almost only on the domain's own share.

Downstream scores are not predictable from the mixture beyond noise. On the held-out random mixtures the linear mixture model reaches $r = 0.52$ for K_{ft} and 0.56 for LISTOPS, but only 0.20 for K_{zs} and 0.18 for R, with RMSE equal to or worse than predicting the mean for K_{zs} and R.

Is loss a proxy for downstream scores? Only weakly. Across mixtures, K_{ft} tracks mean natural-language loss with Spearman $\rho = -0.59$ and K_{zs} with $\rho = -0.44$. The reasoning scores track no loss ($\lvert\rho\rvert \leq 0.39$, and ≤ 0.15 for the composite).

5 Discussion

Reading the hypothesis against the results. The log-linear intuition is right about loss, and it is what scaling laws predict: at a fixed budget a domain's share is its data quantity, and loss falls roughly linearly in log data. The hypothesis goes wrong in two places. The first is the step from loss to downstream performance. The second is the idea of a saturation point. Loss never saturates inside the simplex, and downstream accuracy in this regime is insensitive to share across the whole interior. It moves only when a domain nearly fills the corpus.

Why might downstream scores be flat? We see three explanations, which this experiment cannot fully separate. First, the benefit may arrive very early. If the first 2% of a domain carries half of its value, as it does for loss, then every interior mixture already has enough of everything. The sweeps

Table 6: Web sweep with three seeds per point (mean ± SD across seeds). Loss on the knowledge-bearing domains rises smoothly past 50% web and collapses at 100%; the accuracy measures do not track it.

Web share	K_{zs} (%)	K_{ft} (%)	WIKI loss	ARXIV loss
0	27.47 ± 0.51	46.85 ± 0.90	4.18	3.43
0.25 (uniform)	27.29 ± 0.40	46.87 ± 0.21	4.18	3.49
0.50	27.56 ± 0.25	46.55 ± 0.67	4.20	3.58
0.75	27.32 ± 0.22	47.17 ± 1.15	4.28	3.76
1.00	27.31 ± 0.10	45.65 ± 0.53	5.29	6.09

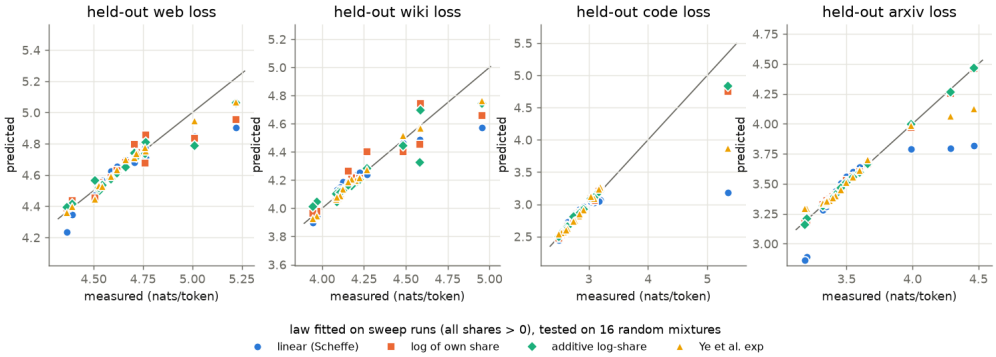


Figure 5: Predicted against observed scores on the 16 held-out random mixtures, for laws fitted on the 29 structured mixtures with every share above zero. Loss is predictable; downstream scores are not predictable beyond noise.

are consistent with downstream saturation well below 20%. However, the models with a domain absent ($p = 0$) also show no drop on the downstream tasks, even though removing a domain raises its loss by 0.9–4.0 nats relative to uniform, which argues against even this reading. Second, the probes may measure general language-model quality, which all mixtures with a majority of natural language share. Pure code, the only mixture with no natural language, is the one clear outlier. Third, the instruments have low sensitivity. Zero-shot accuracy is 2–3 points above chance, and PROOFWRITER has a seed SD of 2.6 points with occasional large jumps: three runs score 68–71% against a typical 56–60%. Effects under about 1 point on knowledge or 2–3 points on reasoning would be invisible with one to three seeds.

Relation to prior work. Our loss results agree with Ye et al. [2025] and Shukor et al. [2025]: loss is a smooth, predictable function of the mixture. A plain log-share form is competitive with the exponential form here and extrapolates better to very small shares. The direction of the code effect in Aryabumi et al. [2025] appears only as a small LISTOPS association. Their gains were measured on models 30–160 times larger and on natural-language reasoning benchmarks that our models cannot attempt zero-shot, so our result does not contradict theirs. RegMix's observation that web text is not harmful [Liu et al., 2025] is consistent with our web sweep. Wikipedia, not web or code, is the most useful single source for our knowledge probes, and it is the only domain whose pure corpus beats the uniform mixture on both knowledge measures.

Failure modes of the instruments. The randomly initialized controls are high. Fine-tuned SCIQ reaches 37.6% with no pre-training against a chance level of 25%, so about 12 points come from answer-string priors that can be learned from the SCIQ training answers. PROOFWRITER reaches 55.9% against a chance level of 33%, leaving only about 4 points that pre-training can explain. ARC-Challenge is below chance (20.0% mean vs. 25%) for every pre-trained model, as expected for a set filtered against co-occurrence baselines.

Arithmetic is bimodal. After 1,500 fine-tuning steps a model either learns 3-digit addition and subtraction (60% of runs score above 85%) or largely fails (20% score below 50%). The three uniform seeds score 8%, 19%, and 97%, and the randomly initialized control scores 76%, so pre-

Table 7: Test RMSE (nats per token) of mixture laws on 16 held-out random mixtures. Parameter counts are in parentheses. Best per row in **bold**.

Held-out loss	Predict the mean	Linear (4)	Log of own share (2)	Additive log-share (5)	Exponential (6)
WEB	0.215	0.100	0.092	0.073	**0.047**
WIKI	0.262	0.110	0.102	0.094	**0.051**
CODE	0.681	0.547	0.153	**0.132**	0.372
ARXIV	0.360	0.238	0.014	**0.011**	0.109

Table 8: Transfer matrix of the additive log-share law (equation 5): change in loss on the row domain, in nats, per e-fold increase in the share of the column domain. Bootstrap SE is at most 0.04. Own-domain effects are in **bold**.

Loss on ↓ / share of →	WEB	WIKI	CODE	ARXIV
WEB	$\mathbf{-0.22}$	-0.08	0.00	0.01
WIKI	-0.08	$\mathbf{-0.29}$	0.01	0.01
CODE	-0.05	-0.04	$\mathbf{-0.46}$	-0.03
ARXIV	-0.01	-0.01	0.00	$\mathbf{-0.26}$

training often hurts this probe. With a seed SD of 23 points we can draw no conclusion about mixtures. Code share correlates negatively with arithmetic ($r = -0.39$ across mixtures); we treat this as a hypothesis only.

5.1 Limitations

Scale. 17M parameters and 60M tokens are four to five orders of magnitude below production pretraining. Reasoning benchmarks and knowledge-intensive tasks in the usual sense are out of reach. Our reasoning measure is fine-tuning transfer to LISTOPS and PROOFWRITER, which is a different construct from zero-shot reasoning. The downstream conclusions should not be extrapolated to large models.

Sensitivity and seeds. The downstream nulls bound effects at roughly 1–3 points; they do not show that the effects are zero. We have three seeds at 14 mixtures and one seed elsewhere. Welch tests with 3 vs. 3 seeds have little power, the uniform seeds are reused in every contrast, and we examined about ninety contrasts without a global correction. Only the very large effects, such as pure code on knowledge, are safe.

Data-dependent choices. We fixed the composites from the uniform pilot and the randomly initialized control before comparing mixtures, but this is still a data-dependent choice; we report individual tasks so that readers can recompute any composite. We chose to fit simplex laws on interior mixtures only after seeing that $p = 0$ is a cliff. We reduced the model and budget from the planned 34M parameters and 100M tokens because the GPUs were shared, and we added 18 replicate runs part-way (Appendix A).

Confounds. Raising one share lowers the others, and the sweeps hold the ratio of the others fixed, not their amounts. At a fixed budget, share and absolute token count are the same variable, so we cannot tell a log law in share from a log law in data quantity.

Coverage. We use one corpus per domain, one architecture, one budget, and one tokenizer. "Code" is Python only and "web" is one C4 shard. Our literature review is at the level of abstracts.

5.2 Broader implications

For practitioners working at small scale, the result is useful in one direction: a mixture can be tuned for loss with a log-share law fitted on about thirty small runs. It is a warning in the other direction: loss differences of this kind did not turn into measurable downstream differences unless a domain was pushed to 75% or more. Loss-level mixing laws are therefore not, on their own, evidence about task performance.

6 Conclusion

We tested four specific claims about how pre-training mixture proportions shape performance, using 81 small language models trained on 53 controlled mixtures of four domains. Held-out loss on a domain follows a log-linear law in that domain's share, with no saturation at 20%, and a law fitted on 29 mixtures predicts loss on unseen mixtures. Downstream accuracy does not follow the hypothesized laws. It is flat within noise across interior mixtures, code share does not correlate with reasoning scores ($r \approx 0$, not > 0.7), and web text above 50% does not measurably hurt knowledge tasks. The one robust downstream effect is that a corpus without natural language is clearly worse on knowledge tasks.

The key takeaway is that, at this scale, a law for loss is not a law for downstream performance.

Several steps follow. The sweep design should be repeated at 100–500M parameters and at least 1B tokens, where zero-shot benchmarks carry signal; the coarse grid $\{0, 0.05, 0.2, 0.5, 1\}$ with three seeds was the informative part of our design. Future runs should use at least five seeds per point and continuous metrics such as log-likelihood margins, so that 0.5-point effects are detectable. Varying the token budget would separate share from the amount of in-domain data and test whether the log-share slopes are budget-invariant. The region between 0% and 2%, where roughly half of each domain's loss benefit is realized, deserves a finer grid. Open questions remain: whether the flatness persists once models can use domain knowledge zero-shot, whether the LISTOPS–code association is a bracket-syntax effect, and why pre-training makes arithmetic harder to learn for some seeds.

References

Viraat Aryabumi, Yixuan Su, Raymond Ma, Adrien Morisot, Ivan Zhang, Acyr Locatelli, Marzieh Fadaee, Ahmet Üstün, and Sara Hooker. To code, or not to code? exploring impact of code in pre-training. In *International Conference on Learning Representations*, 2025. arXiv:2408.10914.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Arman Cohan, Franck Dernoncourt, Doo Soon Kim, Trung Bui, Seokhwan Kim, Walter Chang, and Nazli Goharian. A discourse-aware attention model for abstractive summarization of long documents. In *Proceedings of NAACL-HLT*, 2018.

Simin Fan, Matteo Pagliardini, and Martin Jaggi. DoGE: Domain reweighting with generalization estimation. In *International Conference on Machine Learning*, 2024.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, 2022.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Qian Liu, Xiaosen Zheng, Niklas Muennighoff, Guangtao Zeng, Longxu Dou, Tianyu Pang, Jing Jiang, and Min Lin. RegMix: Data mixture as regression for language model pre-training. In *International Conference on Learning Representations*, 2025. arXiv:2407.01492.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017.

Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *Proceedings of EMNLP*, 2018.

Nikita Nangia and Samuel R. Bowman. ListOps: A diagnostic dataset for latent tree learning. In *Proceedings of the NAACL Student Research Workshop*, 2018.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Mustafa Shukor, Louis Bethune, Dan Busbridge, David Grangier, Enrico Fini, Alaaeldin El-Nouby, and Pierre Ablin. Scaling laws for optimal data mixtures. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2507.09404.

Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of ACL-IJCNLP*, 2021.

Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of NAACL-HLT*, 2019.

Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, 2023.

Johannes Welbl, Nelson F. Liu, and Matt Gardner. Crowdsourcing multiple choice science questions. In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, 2017.

Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems*, 2023.

Jiasheng Ye, Peiju Liu, Tianxiang Sun, Jun Zhan, Yunhua Zhou, and Xipeng Qiu. Data mixing laws: Optimizing data mixtures by predicting language modeling performance. In *International Conference on Learning Representations*, 2025. arXiv:2403.16952.

A Implementation details and deviations from the plan

Compute. We used four NVIDIA RTX A6000 GPUs (48 GB), shared with three other concurrent jobs for most of the session. Pre-training took 11.4 GPU-hours and evaluation 3.8 GPU-hours, with 4–21 minutes of training per run depending on contention. We used no LLM API. Software: Python 3.14.2, PyTorch 2.14.1, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.6, tokenizers 0.23.3.

Deviations from the pre-registered plan. We wrote the plan before training any model and logged the following changes.

- **Smaller model and budget.** The plan called for 34M parameters, 100M tokens, and context 512. Because the GPUs were shared, we switched to 17M parameters, 60M tokens, and context 256 before starting any analyzed run, and discarded the single pilot run in the old configuration.
- **Probes fixed on the pilot.** Using only the uniform model and a randomly initialized control, we set K_{zs} to SCIQ plus ARC-EASY, added the planned fallback K_{ft}, set R to LISTOPS plus PROOFWRITER, and excluded arithmetic (random init 76%, pre-trained 8% in the pilot).
- **Extra replicates.** All 63 planned runs finished. We then added 18 seed replicates (every $p = 0$ and $p = 1$ model, and the code sweep at 0.2 and 0.5). We chose these points before analyzing the full results, but after inspecting 9 runs while debugging the analysis script.
- **Zero-share mixtures excluded from law fitting.** After seeing that $p = 0$ is a cliff that no smooth law fits, we fitted simplex laws on interior mixtures only. We also tried an "effective-share" law, which fitted poorly, and dropped it.
- **Curve-shape criterion.** With nine non-zero points per sweep, the three-parameter models are only loosely identified, so we use leave-one-out error as the primary criterion.
- **Literature.** The automated resource-gathering stage was skipped, so we assembled the corpora, evaluation data, and literature review in-session.